

Kaggle Diabetes Classification

Validation and interpretation with logistic regression and decision tree baselines

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1 Abstract

This article presents a careful binary classification workflow for the Kaggle Pima Indians Diabetes Database. The dataset contains 768 records and a diabetes outcome prevalence of 34.9%. The workflow treats implausible zero values as missing for selected clinical fields, imputes medians from the training set, compares logistic regression and decision tree models, and evaluates performance with a stratified holdout set and 5-fold cross-validation. The selected model is logistic, with holdout ROC-AUC of 0.916, sensitivity of 64.8%, specificity of 94.0%, and F1 of 73.7%. This is a portfolio classification exercise, not a clinical diagnostic tool.

2 Introduction

Diabetes classification is a common ML exercise, but health data require careful wording. A model can be technically interesting without being clinically valid. This project therefore emphasizes validation, interpretability, and limitations.

The research question is:

Which variables help classify diabetes outcome, and how can the model be validated without overstating clinical meaning?

3 Data

Table 1: Dataset summary.

Item	Value
Canonical source	Kaggle Pima Indians Diabetes Database
Records	768
Predictors	8
Outcome prevalence	34.9%
Zero-as-missing fields	Glucose, BloodPressure, SkinThickness, Insulin, BMI

4 Methods

Zeros are converted to missing for clinical fields where zero is biologically implausible. Median imputation is learned only on the training set. Logistic regression uses standardized predictors, making coefficient magnitudes comparable. A decision tree is included as an interpretable nonlinear baseline.

The main validation split is stratified 80/20. Cross-validation uses five stratified folds on the training data.

5 Results

5.1 Holdout Metrics

Table 2: Holdout model metrics at threshold 0.50.

Model	Accuracy	Sensitivity	Specificity	Precision	F1	ROC-AUC
logistic	83.8%	64.8%	94.0%	85.4%	73.7%	0.916

Model	Accuracy	Sensitivity	Specificity	Precision	F1	ROC-AUC
tree	77.3%	72.2%	80.0%	66.1%	69.0%	0.845

5.2 Confusion Matrix

Table 3: Selected-model holdout confusion matrix.

Observed	Predicted	Records
No diabetes	No diabetes	94
No diabetes	Diabetes	6
Diabetes	No diabetes	19
Diabetes	Diabetes	35

5.3 ROC Curve

5.4 Cross-Validation

Table 4: Five-fold cross-validation summary.

Model	Accuracy	Sensitivity	Specificity	F1	Mean ROC-AUC
logistic	74.4%	52.4%	86.2%	59.0%	0.814
tree	72.8%	58.4%	80.5%	59.8%	0.769

5.5 Variable Interpretation

Table 5: Standardized logistic regression coefficients.

Variable	Coefficient	Direction
Glucose	1.081	Higher probability
BMI	0.583	Higher probability
Pregnancies	0.347	Higher probability
DiabetesPedigreeFunction	0.272	Higher probability
Age	0.172	Higher probability
BloodPressure	-0.157	Lower probability
Insulin	-0.078	Lower probability
SkinThickness	0.034	Higher probability

5.6 Calibration

6 Discussion

The selected model provides a useful benchmark for classification and interpretation. Logistic regression is especially appropriate for a portfolio health example because it is transparent: the analyst can inspect direction and relative magnitude of standardized coefficients.

The confusion matrix is as important as ROC-AUC. Depending on whether the intended use is screening, triage, or general risk scoring, the threshold would need to be chosen differently. This project keeps the threshold at 0.50 and reports sensitivity and specificity clearly.

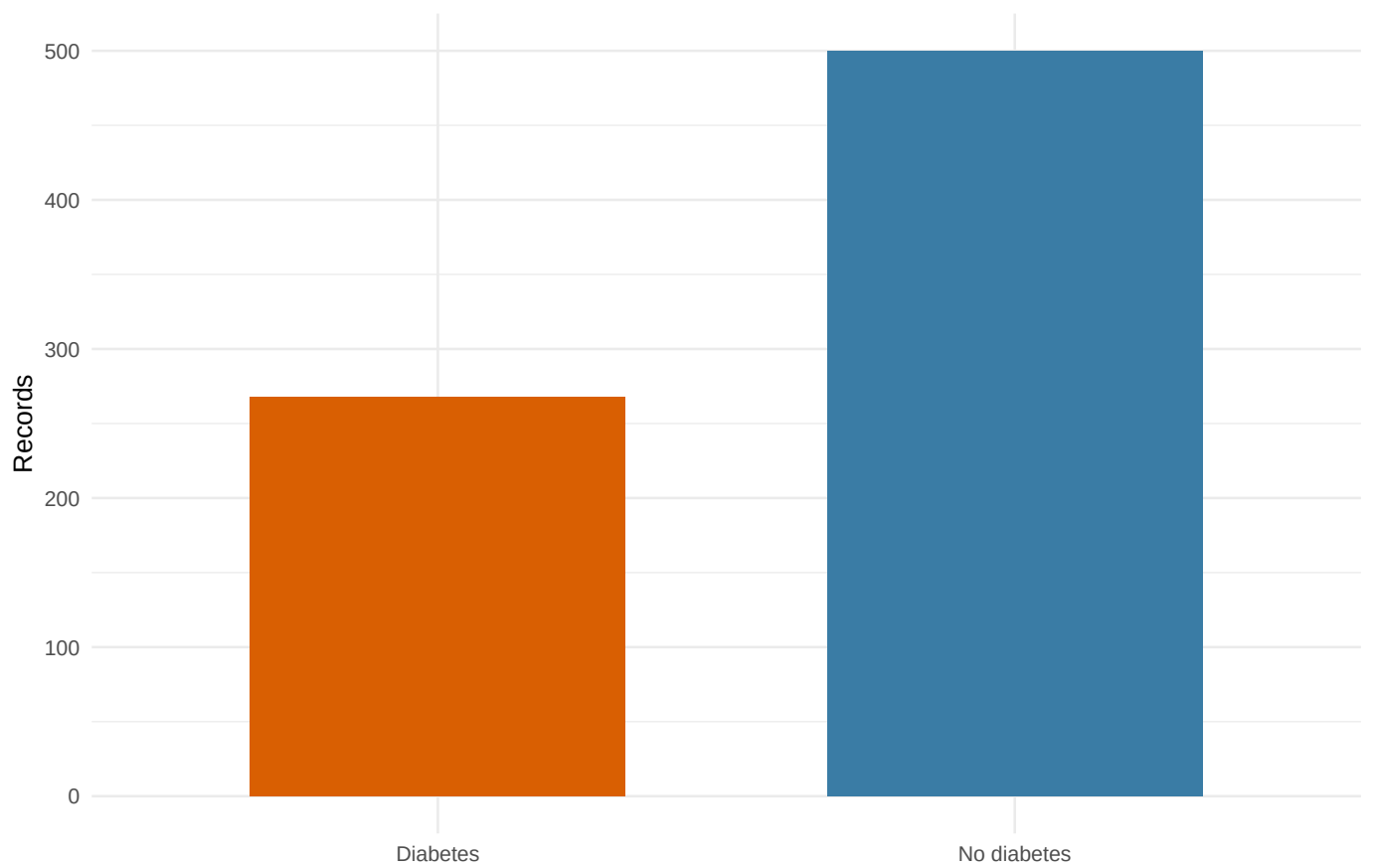


Figure 1: Outcome distribution.

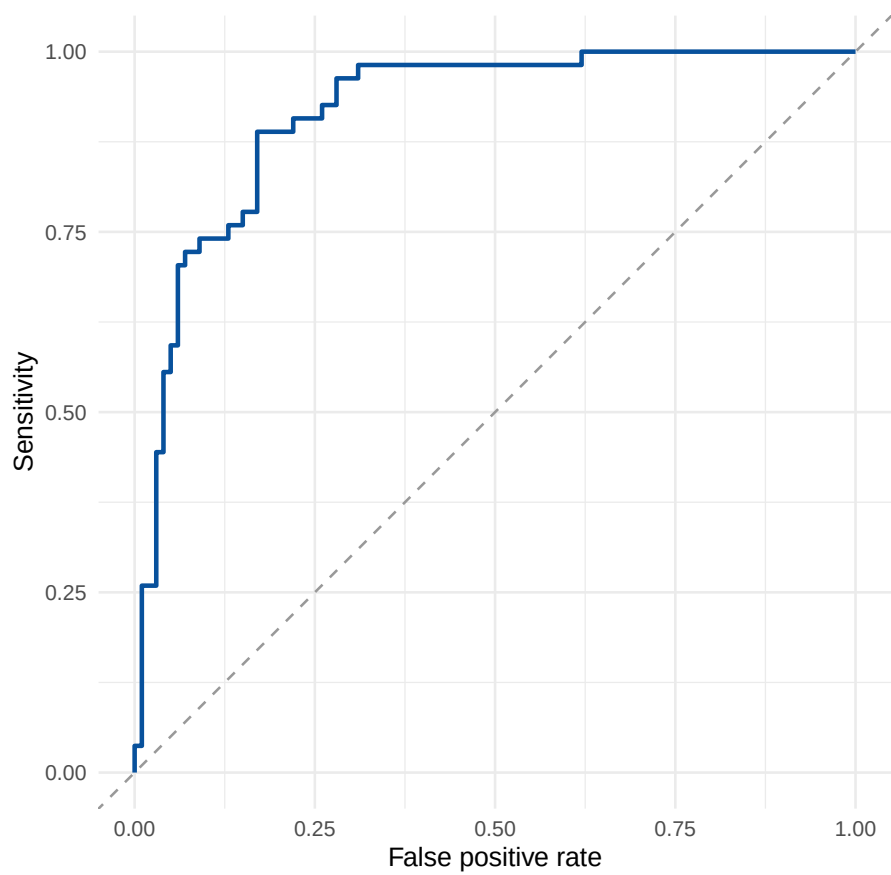


Figure 2: Selected-model ROC curve.

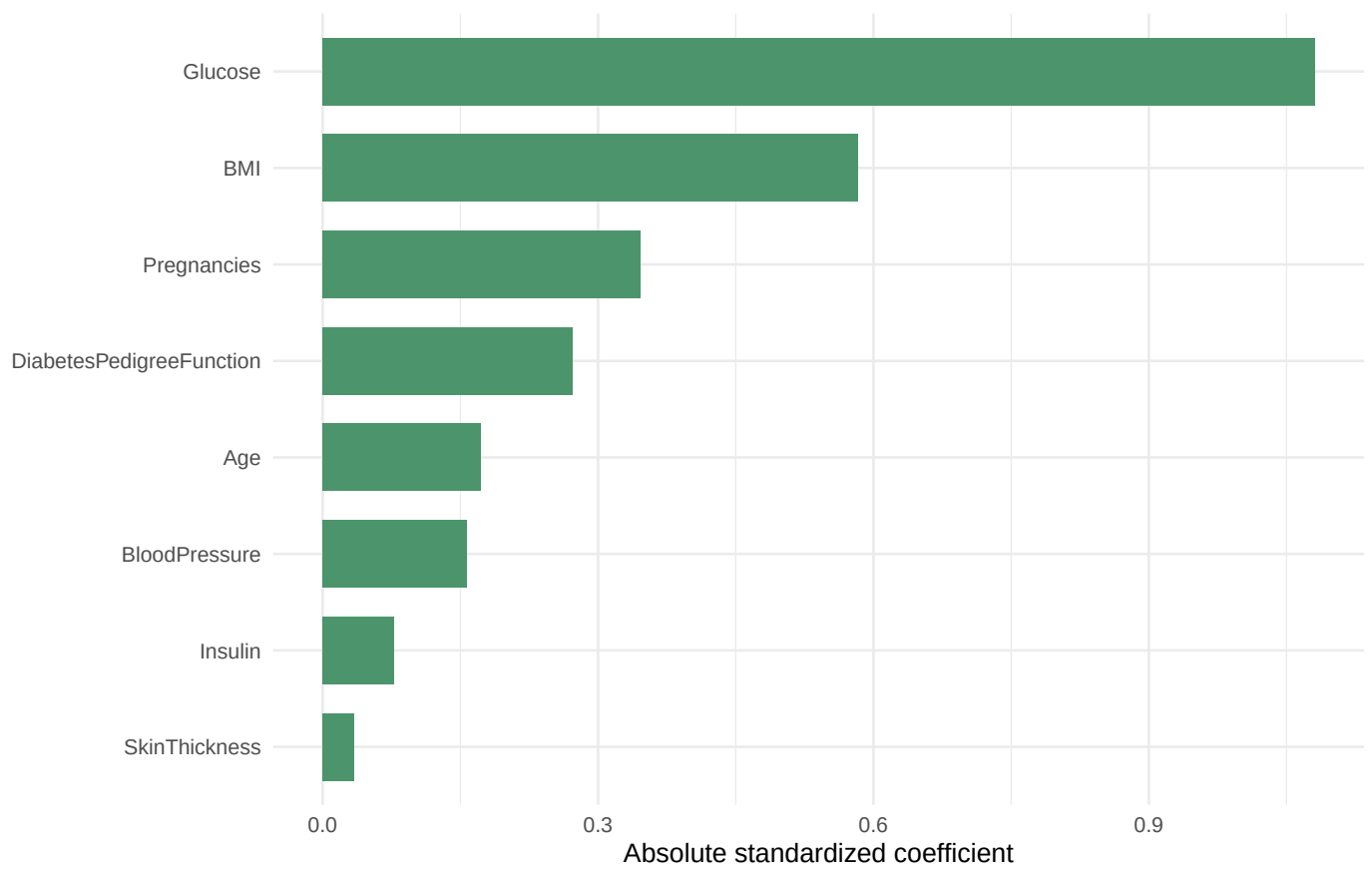


Figure 3: Logistic regression variable importance.

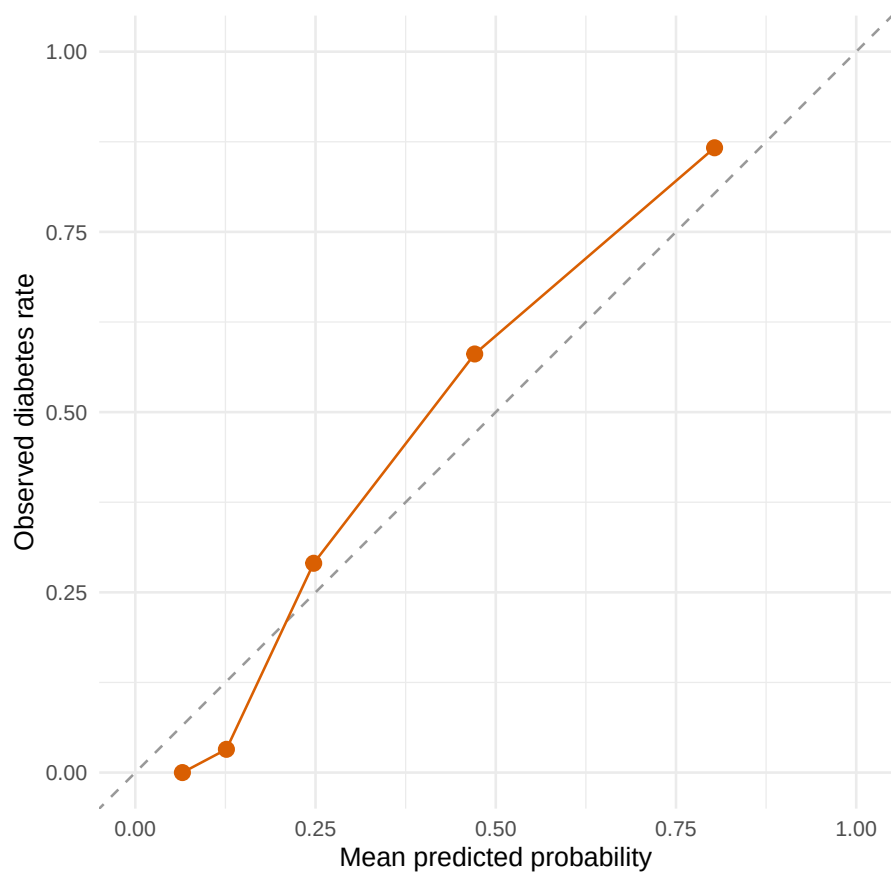


Figure 4: Calibration bins for the selected model.

7 Limitations

This dataset is small, historical, and population-specific. It is not representative of Brazil or of general clinical practice. The model does not include symptoms, medication history, laboratory follow-up, social determinants, or clinician review. The output must not be interpreted as diagnosis or medical advice.

8 Conclusion

S13 delivers a complete, reproducible diabetes classification workflow with careful preprocessing, stratified validation, model comparison, ROC-AUC, calibration, and interpretation. The strongest portfolio signal is not only the model score, but the responsible framing of what the model can and cannot mean.

9 References

Kaggle Pima Indians Diabetes Database. <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>

Public CSV mirror used for local reproducibility. <https://raw.githubusercontent.com/npradaschnor/Pima-Indians-Diabetes-Dataset/master/diabetes.csv>

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